Task 9 — Failure Handling & Resilience

Data Analyst · WEEK 3 · PHASE 2 · Study Guide & Build Pipeline

DATA ANALYST PlaceMux · Phase 2 · Study Guide

1 · Where this fits

This task sits in **WEEK 3 · PHASE 2**. Across the whole team this day is about **Failure Handling & Resilience**. You turn raw events into decisions. A dashboard nobody trusts is worse than no dashboard. Your job is metrics that are correct, sourced, and tied to an actual decision the founder would make.

Your focus this task: Deepen revenue views.

By end of task, the founder should be able to verify: Payment failures handled gracefully and observably.

2 · Learning objectives

By the end of this study guide and task you will be able to:

- Deliver “ARPU + cohort revenue” to a demoable, real-data standard.
- Explain how your work is verified and how it scores out of 100.
- Avoid the specific failure modes that make this kind of work look 'done' when it isn't.

3 · Before you start (prerequisites & inputs)

Make sure you have these in place — missing inputs are the most common reason a task stalls:

- SQL: joins, aggregations, window functions on the product database.
- The difference between an event, a metric and a KPI.
- How analytics events are emitted and where they land.
- Basic data quality checks: nulls, duplicates, suspicious spikes.
- Awareness of consent/DPDP constraints on what you may track.

> **Upstream dependency — confirm this is ready before you begin**
> • Waiting on: Payment data. If it's late, your task slips — chase it early and agree an ETA rather than waiting silently.

4 · Key concepts to understand first

Event → metric → decision

Every metric must trace back to real events and forward to a decision. If no decision changes based on a number, it's a vanity metric — cut it.

Sourcing & trust

For every number you must be able to say exactly where it comes from. 'I'm not sure why it says that' destroys trust in the whole dashboard.

Data really flowing

'Defined' is not 'working'. The events must actually be firing and landing with real (even if small) values before you call it done.

Sanity & freshness

Numbers that look off usually are. Build freshness and sanity checks so a broken pipe is caught by you, not by the founder.

Decision-grade dashboards

A good launch dashboard answers 'is it working and what do we do next', live, with each number explainable on the spot.

5 · Recommended stack & tools

- SQL on the product / analytics database
- An event-tracking pipeline (define → emit → land → query)
- A dashboarding tool / notebook for live views
- Data-quality checks (freshness, nulls, dupes)
- Spreadsheet/CSV export for sharing
- A metric dictionary documenting each number's source

6 · The build pipeline (follow in order)

This is the flow to take the task from nothing to demoable. Work top to bottom; don't skip the test and verify stages — that's where 'looks done' becomes 'is done'.

Stage A — Understand & set up

1. Re-read what good looks like and the definition of done (Section 7) so you build to the target, not past it.
2. Confirm every prerequisite and upstream input above is actually in hand.
3. Pull the agreed sample data / contract and set up a clean branch and working environment.

Stage B — Build: ARPU + cohort revenue

1. Define the metric & its source

Define exactly what “ARPU + cohort revenue” measures, the events it depends on, and the decision it informs.

2. Build the query/event & dashboard

Build the query/event wiring and the dashboard view for “ARPU + cohort revenue”.

3. Validate data is really flowing

Confirm real data is actually flowing (even if small) and add a freshness/sanity check.

4. Tie each number to a decision

For each number, be ready to say where it comes from and what action it would trigger.

Stage C — Integrate, verify & make demoable

1. Wire your deliverable(s) into the end-to-end flow and run the whole journey once, start to finish.
2. Ask them to open the dashboard and explain what each number means and where it comes from.
3. Aim for what 'good' looks like: They open the live dashboard with real (even if small) numbers and explain each one.
4. Prepare a 2-minute live demo on real (even if small) data — not slides, not 'it works'.

7 · Definition of Done

You are done when all of these are true and demoable:

- ARPU/cohort revenue live.
- “ARPU + cohort revenue” is complete, persisted/real, and demoable end-to-end.

8 · How your work is evaluated (scored out of 100)

How it's checked: Ask them to open the dashboard and explain what each number means and where it comes from.

Good looks like: They open the live dashboard with real (even if small) numbers and explain each one.

Scoring parameter	Marks
Core deliverable — “ARPU + cohort revenue” built, working & demoable	50
Real-data quality & correctness (real sample data at scale, not a toy/happy-path)	20
Live verification & evidence (demoed live; real numbers, not claims)	15
Dependency, failure & edge-case handling (errors handled; hand-offs honoured)	15
TOTAL	100

9 · Pitfalls to avoid

Worry if any of these are true of your work

- Metrics are ‘defined’ but nothing is actually flowing yet.
- Numbers look off and they can't explain the source.
- Vanity metrics with no link to a decision you'd actually make.
- We'll add payment failure handling later” — with real money, failure handling IS the feature, not an extra.
- Metrics are 'defined' but no data is actually flowing.
- Numbers look wrong and the source can't be explained.
- Vanity metrics with no link to a real decision.

10 · Hand-off & dependencies

- You hand off: Revenue depth. Make sure the next team can pick it up without you.
- You depended on: Payment data.

11 · Self-check before you submit

If you can answer 'yes, and I can show it live' to each, you're ready:

- Can you show me ‘ARPU + cohort revenue’ working live, rather than just describing it?
- Walk me through what happens if a payment fails halfway — does the student lose money OR the application?
- How do we know our records match exactly what the gateway says we collected?
- Are we in real-money mode or still test mode — and what's left before real money?

12 · Go deeper (optional further study)

- Designing an event taxonomy
- Funnel and cohort analysis
- Data-quality monitoring and freshness SLAs
- Defining North-Star and guardrail metrics
- Privacy-aware analytics under DPDP